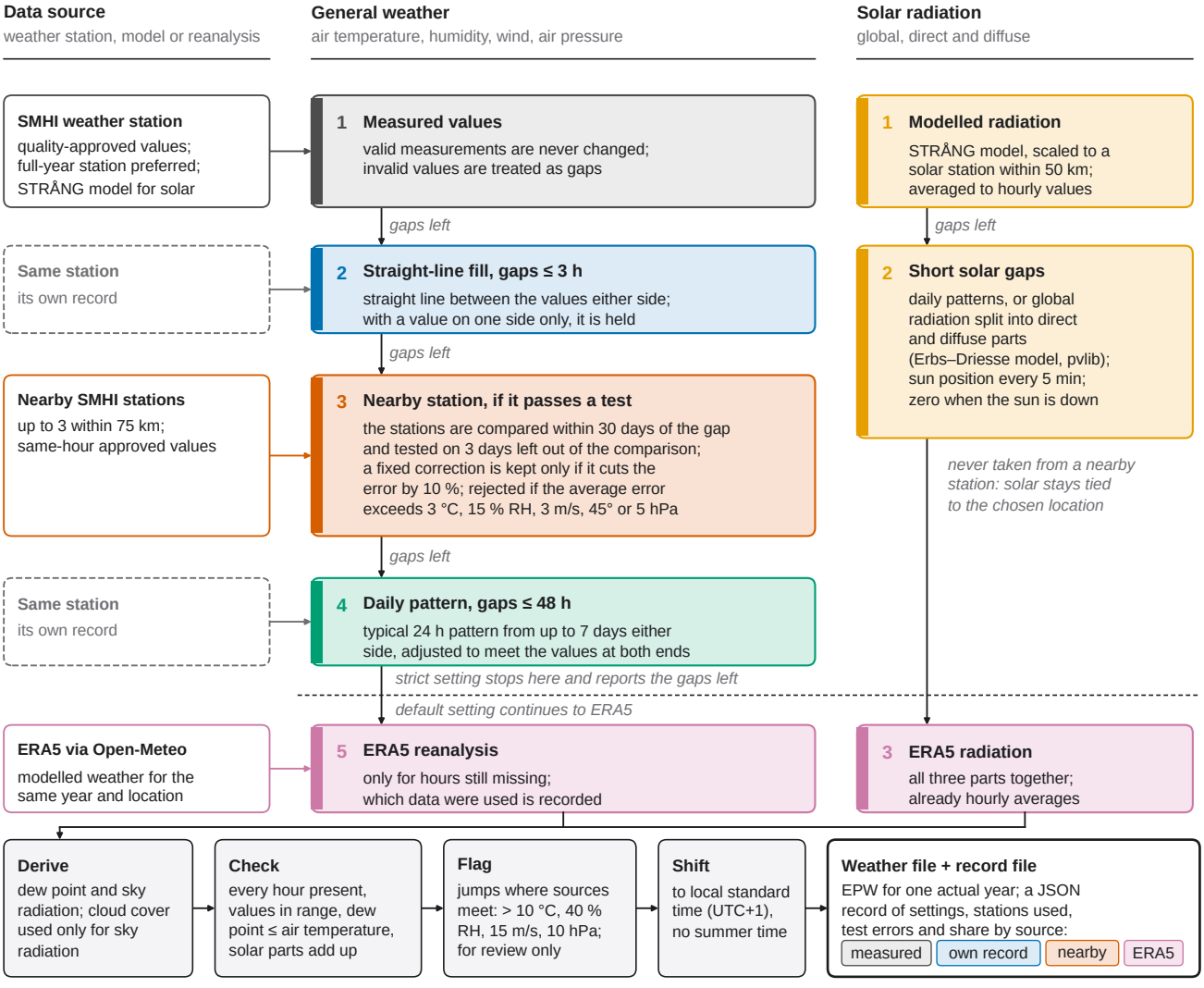
Data source
weather station, model or reanalysis
General weather
air temperature, humidity, wind, air pressure
Solar radiation
global, direct and diffuse
SMHI weather station
quality-approved values;
full-year station preferred;
STRÅNG model for solar
1 Measured values
valid measurements are never changed;
invalid values are treated as gaps
1 Modelled radiation
STRÅNG model, scaled to a
solar station within 50 km;
averaged to hourly values
gaps left
gaps left
Same station
its own record
2 Straight-line fill, gaps ≤ 3 h
straight line between the values either side;
with a value on one side only, it is held
2 Short solar gaps
daily patterns, or global
radiation split into direct
and diffuse parts
(Erbs–Driesse model, pvlib);
sun position every 5 min;
zero when the sun is down
gaps left
Nearby SMHI stations
up to 3 within 75 km;
same-hour approved values
3 Nearby station, if it passes a test
the stations are compared within 30 days of the gap
and tested on 3 days left out of the comparison;
a fixed correction is kept only if it cuts the
error by 10 %; rejected if the average error
exceeds 3 °C, 15 % RH, 3 m/s, 45° or 5 hPa
never taken from a nearby
station: solar stays tied
to the chosen location
gaps left
Same station
its own record
4 Daily pattern, gaps ≤ 48 h
typical 24 h pattern from up to 7 days either
side, adjusted to meet the values at both ends
strict setting stops here and reports the gaps left
default setting continues to ERA5
ERA5 via Open-Meteo
modelled weather for the
same year and location
5 ERA5 reanalysis
only for hours still missing;
which data were used is recorded
3 ERA5 radiation
all three parts together;
already hourly averages
Derive
dew point and sky
radiation; cloud cover
used only for sky
radiation
Check
every hour present,
values in range, dew
point ≤ air temperature,
solar parts add up
Flag
jumps where sources
meet: > 10 °C, 40 %
RH, 15 m/s, 10 hPa;
for review only
Shift
to local standard
time (UTC+1),
no summer time
Weather file + record file
EPW for one actual year; a JSON
record of settings, stations used,
test errors and share by source:
measured
own record
nearby
ERA5